

Nikhilesh Vijay Chirutala

Backend Software Engineer

+91-9618873598 • nikhilesh.chirutala@gmail.com • [Linkedin](#) • [Github](#) • [Portfolio](#) • Bengaluru, IN

SUMMARY

Backend Software Engineer with 3+ years of experience designing, developing, and maintaining production applications using Python, FastAPI, REST APIs, MongoDB, PostgreSQL, MySQL, and AWS. Experienced in backend service development, API design, business logic, data processing, cloud storage workflows, third-party integrations, authentication, authorization, and production troubleshooting. Currently developing Python services for a high-volume cloud contact-center platform, working on call-recording processing, recovery, storage, and service integrations. Also experienced in RAG-based applications, LLM integration, CI/CD, observability, and production support.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), TypeScript

Backend & Frameworks: FastAPI, Django, Flask, REST APIs, API Development, Backend Automation

Databases: MongoDB, PostgreSQL, MySQL, SQL, Neon DB

Cloud & AWS: AWS, Amazon S3, AWS Lambda, Vercel

Architecture & Integration: Microservices, WebSockets, API Integration, Third-Party Integrations

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Security: Authentication, Authorization, RBAC, ACL, JWT

DevOps & Observability: GitHub Actions, CI/CD, Linux, Datadog, Grafana, Sentry

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

Frontend: React.js, Next.js, Angular, Jinja2, Tailwind CSS

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog

- Develop and maintain production-grade Python services for a high-volume cloud contact-center platform, supporting call-recording ETL, data processing, storage workflows, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, supporting feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Built AWS S3 audit and recovery workflows for recording storage, including deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry for monitoring service health, tracking errors, diagnosing processing failures, and supporting production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing troubleshooting and implementing fixes to maintain reliable recording-processing workflows.

Software Engineer • Mantra Technologies

Dec 2023 – Mar 2026 | Hyderabad, India

- Developed backend services, REST APIs, business logic, and data-driven workflows using Python and FastAPI across multiple production applications.
- Designed and implemented API-driven application functionality, connecting backend services, databases, external integrations, and application workflows.
- Implemented authentication, authorization, and RBAC/ACL-based access-control workflows across application resources.
- Integrated external services and backend components, including communication, storage, notification, and application integrations.
- Supported production releases and resolved application issues through API testing, debugging, root-cause analysis, deployment validation, and defect resolution.

PERSONAL PROJECTS

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.

- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

[AnyScrum](#) — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

[JS Studio](#) — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

[Guider](#) — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur